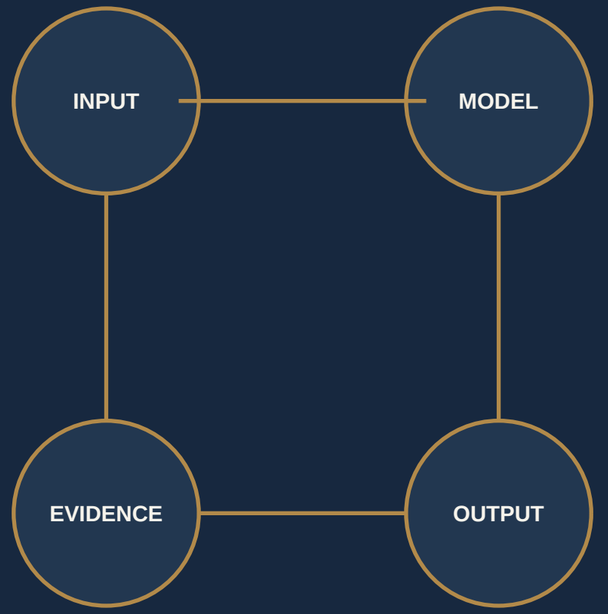
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robot Learning

AI.403 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define learning from demonstrations.
- Apply key methods including imitation learning.
- Build or analyze a practical system involving reinforcement learning.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Learning from demonstrations

02

METHODS

Imitation learning

03

SYSTEMS

Reinforcement learning

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Learning from demonstration

Learning from demonstrations

02 · Imitation learning

Imitation learning

03 · Reinforcement learning

Reinforcement learning

04 · Self-supervised learning

Self-supervised learning

05 · Adaptive behavior and learnir

Adaptive behavior and learning-based control

06 · Mandatory practical: Train a r

Mandatory practical: Train a robot to perform a task through demonstration or reinforcement learning

Core ideas, made explicit

AI.403 • Distinguish definitions, assumptions and evidence

- Learning from demonstrations
- Imitation learning
- Reinforcement learning
- Self-supervised learning
- Adaptive behavior and learning-based control

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.403 • Distinguish definitions, assumptions and evidence

- Mandatory practical: Train a robot to perform a task through demonstration or reinforcement learning

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM

Learning from demonstrations

INPUTS

Imitation learning

METHOD

Reinforcement learning

OUTPUT

Self-supervised learning

EVALUATION

Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving learning from demonstrations.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using learning from demonstrations.
- Implement or simulate a baseline using imitation learning.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.